

ACTION CHUNKING & TEMPORAL ENSEMBLING

Emitting Horizon $H=8$ Action Blocks with Exponential Overlap Weighting for Smooth Execution

Chunk 1 ($t = 0$ ms):



Chunk 2 ($t = 50$ ms):



TEMPORAL OVERLAP BLEND

Ensembled u_t :



MATHEMATICAL FORMULATION: $u_t = \sum w_i \cdot a_t^{(i)} / \sum w_i$ where $w_i = \exp(-m \cdot (t - t_{\text{start}}))$

Recent predictions carry higher exponential weights; older tail steps decay gracefully to eliminate joint torque jerks.

THE ACTION CHUNKING PRINCIPLE

Chunking amortizes high inference latency over future time steps; temporal ensembling eliminates inter-chunk torque discontinuities.